

Total No. of Questions : 8]

SEAT No. :

PD-4063

[Total No. of Pages : 3

[6402]-22

S.E.

(Electronics/E&TC/Electronics (VLSI & Design & Technology) Electronics & Communication Advanced Communication Technology)

ELECTRONIC CIRCUITS

(204181) (2019 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary

Q1) a) Enlist the advantages and disadvantages of SMPS over linear power supply. Draw the block diagram of SMPS and explain each block in detail. [8]

b) Design an adjustable voltage regulator using LM317 for output voltage 5 V to 15 V and draw necessary connection diagrams. Assume $R_1 = 240 \Omega$, $I_{adj} = 100 \mu A$. [9]

OR

Q2) a) Draw the block diagram of a regulated DC power supply and explain the function of each block in detail. [9]

b) Explain the following performance parameters of DC power supply [8]

- i) Load regulation
- ii) Line regulation,
- iii) Ripple rejection and
- iv) Efficiency.

P.T.O.

Q3) a) Explain the following op-amp parameters: [8]

- i) Input offset Voltage
 - ii) CMRR
 - iii) Slew Rate
 - iv) Unity gain bandwidth.
- b) What is current mirror circuit? Draw and explain working of differential amplifier with current mirror circuit. Why current mirror circuit is used in Op Amp? [6]
- c) State the values of all ideal parameters of Op Amp. [3]

OR

Q4) a) Draw block diagram of Op-Amp. Explain the functions of each block in detail. [8]

- b) A dual input, balanced output (DIBO) differential amplifier has following specifications: $R_{C1} = R_{C2} = 2.2 \text{ k}\Omega$, $R_E = 4.7 \text{ k}\Omega$, $R_{in1} = R_{in2} = 50 \Omega$, $+V_{CC} = 10\text{V}$, $-V_{EE} = -10\text{V}$, $\beta_{dc} = \beta_{ac} = 100$ and $V_{BE} = 0.7 \text{ V}$. [6]

Calculate : i) I_C ii) V_{CEQ} iii) Voltage gain : A_d
iv) Input and output resistance

- c) Explain how to improve CMMR in Op Amp. [3]

Q5) a) What are the limitations of ideal integrator? How these are overcome in practical integrator? Draw the circuit diagram of practical integrator and its frequency response. Write equation for output voltage V_o and explain its operation. [9]

- b) Draw an inverting summing amplifier with three inputs. Derive an expression for its output voltage. [6]

- c) Draw square wave generator using Op Amp with relevant waveforms [3]

OR

- Q6)** a) Explain with a neat circuit diagram, working of inverting symmetric Schmitt trigger with its input output waveform and hysteresis plot. [9]
- b) Enlist the requirements of instrumentation amplifier. Draw and explain instrumentation amplifier using three Op Amp. [6]
- c) Draw full wave precision rectifier with relevant waveforms. [3]
- Q7)** a) Draw circuit diagram and explain successive approximation ADC. [6]
- b) Explain the following characteristics of DAC [6]
- i) Resolution, ii) Accuracy and iii) Settling time.
- c) Draw the block diagram of PLL and explain each block in detail. [6]

OR

- Q8)** a) Draw and explain circuit diagram for 3-bit weighted resistor D/A converter, and give output voltage equation for the same. [6]
- b) Calculate the output frequency f_o , lock range and capture range of PLL if $R_T = 10 \text{ k}\Omega$, C_T (Timing capacitor) = $0.01 \mu\text{F}$, C_F (Filter capacitor) $10 \mu\text{F}$, R (Internal filter resistor) $3.6 \text{ k}\Omega$. [6]
- c) Draw and explain the block diagram of frequency multiplier with relevant output waveforms. [6]

